

Turing

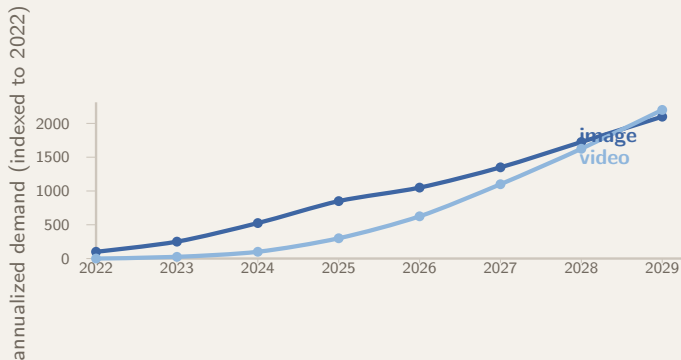
Diffusion Processing Unit

The silicon wedge for always-on generative inference

Stable Diffusion made image generation feel ordinary almost overnight. The next question is how to serve that demand—and the wave of video that follows—without paying GPU economics. Open models move generative media from demo to daily use.



Demand is already here, and it is widening



Demand snapshot

2026E	image	1,050
	video	625
2029E	image	2,100
	video	2,200

Image first, video close behind

- ▶ Text-to-image proved the commercial habit. Video compounds the same diffusion logic across many more frames.
- ▶ This is an image market now, and a video market soon after.
- ▶ Once usage is continuous, the buying question is simple: output, margin, and power.

What customers feel

\$15–40 per 1,000 diffusion inferences is tolerable for experimentation and painful for scaled deployment.

What specialized silicon offers

A 5–20× lower inference-cost target and a 10–50× power advantage on the denoising path.

Why a DPU wins this wedge

Fixed loop, not general theater

Diffusion inference repeats a denoising pattern often enough to justify hardware that is priced for the loop, not for every possible model path.

Tile first, system second

A narrow signed-int8 MAC tile already maps to convolution, attention, and MLP-heavy denoising kernels.

Lower variance matters

Paid generation services care about predictable latency, stable thermals, and more images per rack as much as headline TOPS.



Market capture is a wedge story, not a winner-take-all story



AI inference accelerator market
\$25B–40B



Diffusion-heavy deployment wedge
\$3B–8B



Early meaningful capture
\$50M–250M+

This is a focused play for the part of inference where image and video generation are persistent enough to justify specialized economics.

Competition is broad. The advantage is narrow and commercial.

Cloud GPUs

Great default.
Expensive at scale.

Hyperscaler silicon

Powerful, but built for
broad fleets.

AI chip startups

Many chase general AI.
Few stay on diffusion.

Turing: better economics for a repeating diffusion workload, with a cleaner path to trustworthy silicon.

Turing's edge: formal verification is a tape-out advantage

Start small, prove early

The repository begins with a formally tractable tile: signed int8 arithmetic, saturating accumulation, and explicit valid/ready boundaries.

Cut late surprises

Protocol bugs and arithmetic edge cases are cheap in RTL and expensive in signoff. Proof moves that pain earlier.

Sell trust, not only speed

Operators buy confidence that the board, module, and software path will integrate without a long tail of avoidable defects.



Go-to-market: partner with Southern U.S. AI buildouts

- ▶ New AI capacity is easier to shape than installed GPU fleets are to displace.
- ▶ Southern buildouts offer power, land, and cooling headroom.
- ▶ The pitch is simple: more output per watt, per rack, and per megawatt.

Pod pilot

20 MW pod, trial design partner

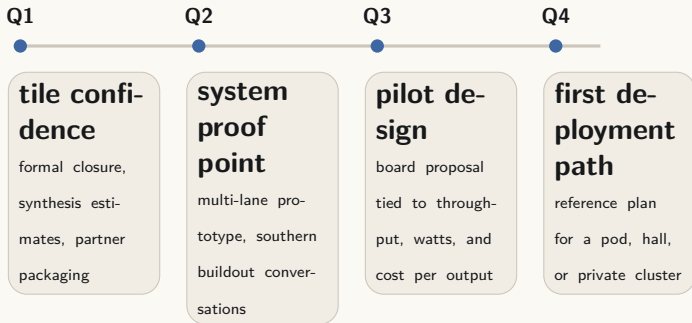
Target profiles, not yet named deals

Texas, Tennessee, and Georgia are attractive because new AI halls are often planned in 20–100+ MW increments, where accelerator choice still affects the economics of the build.

Regional AI hall

50 MW cluster, rack-scale proof

The next 12 months



The first win is not a global platform. It is one buildout where image and video generation are expensive enough to justify a better chip.

The market is already measured in billions.

Image generation made diffusion ordinary; video is widening the same demand curve.

Current market

\$25B–40B in AI inference accelerators, with a \$3B–8B diffusion-heavy wedge already visible.

Current usage

100M generations per month implies roughly \$24M in annualized savings when output costs move from \$25 to \$5 per 1,000.

Projected market

At 250M–500M generations per month, annualized savings rise to roughly \$60M–120M, before video widens the wedge further.